

A - Fizz

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 100 points

Problem Statement

You are given a positive integer N .

Output N lines. The i -th line ($1 \leq i \leq N$) should contain `Fizz` if i is a multiple of 3 , and i if i is not a multiple of 3 .

Constraints

- N is an integer between 1 and 100 , inclusive.

Input

The input is given from Standard Input in the following format:

N

Output

Output the answer.

Sample Input 1

4

Sample Output 1

```
1
2
Fizz
4
```

The first line should contain **1** since **1** is not a multiple of **3**.

The second line should contain **2** since **2** is not a multiple of **3**.

The third line should contain **Fizz** since **3** is a multiple of **3**.

The fourth line should contain **4** since **4** is not a multiple of **3**.

Sample Input 2

```
10
```

Sample Output 2

```
1
2
Fizz
4
5
Fizz
7
8
Fizz
10
```

B - Monocolor

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 200 points

Problem Statement

There are N balls. Each ball is painted in one of N colors: color 1 through color N . The color of the i -th ball ($1 \leq i \leq N$) is C_i .

In one operation, you can change the color of any one ball to any of the N colors.

Find the minimum number of operations required to make all the balls the same color.

Constraints

- $1 \leq N \leq 100$
 - $1 \leq C_i \leq N$
 - All input values are integers.
-

Input

The input is given from Standard Input in the following format:

```
 $N$   
 $C_1$   $C_2$   $\dots$   $C_N$ 
```

Output

Output the answer.

Sample Input 1

```
4  
3 1 2 1
```

Sample Output 1

```
2
```

By changing the color of the first ball to 1 and the color of the third ball to 1, you can make all the balls the same color.

It is impossible to make all the balls the same color by changing the colors of fewer than two balls, so output 2.

Sample Input 2

```
5
3 3 3 3 3
```

Sample Output 2

```
0
```

All balls may have the same color from the beginning.

Sample Input 3

```
9
4 2 3 3 4 1 2 7 1
```

Sample Output 3

```
7
```

C - Inc, Dec, Xor

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 300 points

Problem Statement

There is a length- N integer sequence $A = (A_1, A_2, \dots, A_N)$. Initially, all elements of A are 0.

You will be given Q queries, which should be processed in order. There are two types of queries, each given in one of the following formats:

- 1 x : Increase the value of A_x by 1.
- 2: For each $i = 1, 2, \dots, N$, if $A_i \geq 1$, decrease the value of A_i by 1.

Find the bitwise XOR of $A_1, A_2, \dots, A_N$ immediately after processing each query.

► What is bitwise XOR?

Constraints

- $1 \leq N \leq 5 \times 10^5$
- $1 \leq Q \leq 5 \times 10^5$
- $1 \leq x \leq N$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N Q
query1
query2
⋮
queryQ
```

Each query is given in one of the following 2 formats:

1 x

2

Output

Output Q lines.

The i -th line ($1 \leq i \leq Q$) should contain the bitwise XOR of $A_1, A_2, \dots, A_N$ for A immediately after processing the i -th query.

Sample Input 1

```
2 5
1 2
1 2
1 1
2
2
```

Sample Output 1

```
1
2
3
1
0
```

After processing the first query, $A = (0, 1)$. The bitwise XOR of 0, 1 is 1, so output 1 on the first line.

After processing the second query, $A = (0, 2)$. The bitwise XOR of 0, 2 is 2, so output 2 on the second line.

After processing the third query, $A = (1, 2)$. The bitwise XOR of 1, 2 is 3, so output 3 on the third line.

After processing the fourth query, $A = (0, 1)$. The bitwise XOR of 0, 1 is 1, so output 1 on the fourth line.

After processing the fifth query, $A = (0, 0)$. The bitwise XOR of 0, 0 is 0, so output 0 on the fifth line.

Sample Input 2

```
3 8
1 2
1 3
1 1
1 2
1 1
2
1 3
1 1
```

Sample Output 2

```
1
0
1
2
1
0
1
2
```

D - Inverse and Swap

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 400 points

Problem Statement

You are given a permutation $P = (P_1, \dots, P_N)$ of $(1, \dots, N)$.

Process Q queries in order. There are two types of queries as follows:

- 1 $\times$ y : Swap the values of P_x and P_y .
- 2: Construct the permutation $P' = (P'_1, \dots, P'_N)$ of $(1, \dots, N)$ satisfying the following condition, and replace the values of $P_1, \dots, P_N$ with $P'_1, \dots, P'_N$, respectively. (One can prove that such P' uniquely exists.)
 - $P_{P'_i} = i$ for every integer i satisfying $1 \leq i \leq N$.

Output the values of $P_1, \dots, P_N$ after processing all queries.

Constraints

- $2 \leq N \leq 5 \times 10^5$
 - $1 \leq Q \leq 5 \times 10^5$
 - $(P_1, \dots, P_N)$ is a permutation of $(1, \dots, N)$.
 - $1 \leq x < y \leq N$ for queries of type 1.
 - All input values are integers.
-

Input

The input is given from Standard Input in the following format:

```

 $N$   $Q$ 
 $P_1$   $P_2$   $\dots$   $P_N$ 
query1
⋮
query $Q$ 

```

Here, query _{q} represents the q -th query, and is given in one of the following two formats:

```
1  $x$   $y$ 
```

```
2
```

Output

Output the values of $P_1, \dots, P_N$ after processing all queries, separated by spaces, on one line.

Sample Input 1

```

5 5
2 1 3 5 4
1 2 4
2
1 2 3
1 3 4
2

```

Sample Output 1

```
4 5 2 1 3
```

At the point when each query has been processed, the values of $P_1, \dots, P_N$ are as follows:

- After processing the first query, $P = (2, 5, 3, 1, 4)$.
- After processing the second query, $P = (4, 1, 3, 5, 2)$.
- After processing the third query, $P = (4, 3, 1, 5, 2)$.
- After processing the fourth query, $P = (4, 3, 5, 1, 2)$.
- After processing the fifth query, $P = (4, 5, 2, 1, 3)$.

Sample Input 2

```
7 4
3 7 5 6 4 2 1
2
2
2
2
```

Sample Output 2

```
3 7 5 6 4 2 1
```

Sample Input 3

```
10 8
7 3 2 4 8 5 10 9 1 6
2
1 4 10
1 6 9
2
1 9 10
1 3 10
2
1 4 6
```

Sample Output 3

```
3 10 2 8 6 7 1 5 9 4
```

E - Concentration

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 450 points

Problem Statement

Takahashi is playing a solitaire game similar to the memory-matching game.

There are $2N$ cards. A number is written on the front of each card, and nothing is written on the back. For each i satisfying $1 \leq i \leq N$, there are exactly two cards on which A_i is written. (The A_i are pairwise distinct.)

Takahashi plays the game with these cards using the following procedure.

- Shuffle the $2N$ cards and lay them out face down.
- Set **life** to L and **score** to 0.
- Repeat the following until life becomes 0 or there are no cards left on the table:
 - Choose one face-down card on the table, turn it face up, and check the number X written on it.
 - Choose one face-down card on the table, turn it face up, and check the number Y written on it.
 - If $X = Y$, remove those two cards from the table, and increase the score by X .
 - If $X \neq Y$, turn the two cards face down again, and decrease life by 1.

Find the expected value of the score at the end of the game when Takahashi acts optimally to maximize the score at the end of the game.

Below is a more formal description of the game.

- Takahashi knows the rules of this game described below.
- Takahashi knows the values of $A_1, \dots, A_N$.
- Let B be a sequence obtained by choosing a permutation of the length- $2N$ sequence $(A_1, A_1, A_2, A_2, \dots, A_N, A_N)$ uniformly at random.
- Initially, Takahashi knows none of the values of B . Once he learns the value of B_i , he remembers it completely thereafter.
- Let life be L , score be 0, and S be $\{1, 2, 3, \dots, 2N\}$. Takahashi always knows these values.
- Repeat the following until life becomes 0 or S becomes empty:
 - Based on the information obtained up to this point, Takahashi chooses an element from S , and calls it i .
 - The value of B_i is revealed, and Takahashi learns it.
 - Based on the information obtained up to this point (including the value of B_i), Takahashi chooses an element from $S \setminus \{i\}$, and calls it j .
 - The value of B_j is revealed, and Takahashi learns it.
 - If $B_i = B_j$, remove i and j from S , and add B_i to the score.
 - If $B_i \neq B_j$, decrease life by 1.
- Takahashi acts optimally to maximize the expected value of the score at the end of the game.

Constraints

- $1 \leq N \leq 200$
- $1 \leq L \leq 200$
- $1 \leq A_1 < A_2 < \dots < A_N \leq 10^5$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
 $N$   $L$   
 $A_1$   $A_2$   $\dots$   $A_N$ 
```

Output

Output the answer.

Your output will be considered correct if its absolute or relative error from the true answer is at most 10^{-5} .

Sample Input 1

```
3 2  
1 2 3
```

Sample Output 1

```
3.8666666667
```

The game may proceed as follows, for example. To distinguish the six cards, let us call them A, B, C, D, E, F.

- Start the game with life 2 and score 0.
- Turn card A face up. 3 is written on it.
- Turn card B face up. 2 is written on it.
- Since different numbers are written, turn both cards face down again, and decrease life by 1 to 1.
- Turn card C face up. 3 is written on it.
- Turn card A face up. 3 is written on it.
- Since the same number is written, remove both cards from the table, and increase the score by 3 to 3.
- Turn card D face up. 1 is written on it.
- Turn card E face up. 2 is written on it.
- Since different numbers are written, turn both cards face down again, and decrease life by 1 to 0.
- Since life has become 0, the game ends. The score is 3.

Note that, immediately after turning card C face up in this sequence, Takahashi can make the choice of "turning face up card A, the other already-known card with 3 written on it, based on the fact that 3 was written on the front of card C."

Sample Input 2

```
5 2
2 3 5 7 101
```

Sample Output 2

```
17.8560846561
```

Sample Input 3

```
20 10
10 20 30 40 50 60 70 80 90 100 110 120 130 140 150 160 170 180 190 200
```

Sample Output 3

```
770.7122293087
```


F - Googol Swaps

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 500 points

Problem Statement

You are given a string S of length N consisting of lowercase English letters.

Find the number, modulo 998244353, of strings that S can become after performing the following operation **exactly** 10^{100} times.

- Choose an integer i between 1 and M , inclusive, and swap the A_i -th and B_i -th characters of S .

Constraints

- N and M are integers.
- $2 \leq N \leq 2 \times 10^5$
- $1 \leq M \leq 2 \times 10^5$
- S is a string of length N consisting of lowercase English letters.
- A_i and B_i are integers.
- $1 \leq A_i < B_i \leq N$
- $(A_1, B_1), \dots, (A_M, B_M)$ are pairwise distinct.

Input

The input is given from Standard Input in the following format:

```
 $N$   $M$   
 $S$   
 $A_1$   $B_1$   
 $\vdots$   
 $A_M$   $B_M$ 
```

Output

Output the answer.

Sample Input 1

```
5 3
miria
1 3
2 5
4 5
```

Sample Output 1

```
6
```

The following six strings are possible as the final S :

- marii
- mirai
- miria
- ramii
- rimai
- rimia

Sample Input 2

```
6 6
yiwayi
1 2
1 3
2 3
4 5
4 6
5 6
```

Sample Output 2

```
18
```

Sample Input 3

```
29 25
hexakosioihexekontahexaphobia
1 2
1 4
1 6
1 8
1 15
1 16
2 3
3 4
4 20
5 6
5 8
8 22
8 23
9 15
9 17
11 21
12 20
13 19
14 29
15 28
16 17
18 19
18 21
19 20
20 21
```

Sample Output 3

```
346192062
```

G - ΣΣX

Time Limit: 3 sec / Memory Limit: 1024 MiB

Score : 600 points

Problem Statement

You are given a length- N sequence of non-negative integers $A = (A_1, \dots, A_N)$.

Find the sum of $\text{mex}(A_l, \dots, A_r)$ over all pairs of integers (l, r) satisfying $1 \leq l \leq r \leq N$.

Here, $\text{mex}(A_l, \dots, A_r)$ denotes the smallest non-negative integer not contained in $A_l, \dots, A_r$.

Constraints

- $1 \leq N \leq 3 \times 10^5$
- $0 \leq A_i \leq N$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
N
A_1 A_2 ... A_N
```

Output

Output the answer.

Sample Input 1

```
3
1 2 0
```

Sample Output 1

5

The answer is 5, the sum of the following 6 values.

- $\text{mex}(A_1) = \text{mex}(1) = 0$
- $\text{mex}(A_1, A_2) = \text{mex}(1, 2) = 0$
- $\text{mex}(A_1, A_2, A_3) = \text{mex}(1, 2, 0) = 3$
- $\text{mex}(A_2) = \text{mex}(2) = 0$
- $\text{mex}(A_2, A_3) = \text{mex}(2, 0) = 1$
- $\text{mex}(A_3) = \text{mex}(0) = 1$

Sample Input 2

6
2 1 0 2 1 4

Sample Output 2

31

The answer is 31, the sum of the following 21 values.

- $\text{mex}(A_1) = \text{mex}(2) = 0$
- $\text{mex}(A_1, A_2) = \text{mex}(2, 1) = 0$
- $\text{mex}(A_1, A_2, A_3) = \text{mex}(2, 1, 0) = 3$
- $\text{mex}(A_1, A_2, A_3, A_4) = \text{mex}(2, 1, 0, 2) = 3$
- $\text{mex}(A_1, A_2, A_3, A_4, A_5) = \text{mex}(2, 1, 0, 2, 1) = 3$
- $\text{mex}(A_1, A_2, A_3, A_4, A_5, A_6) = \text{mex}(2, 1, 0, 2, 1, 4) = 3$
- $\text{mex}(A_2) = \text{mex}(1) = 0$
- $\text{mex}(A_2, A_3) = \text{mex}(1, 0) = 2$
- $\text{mex}(A_2, A_3, A_4) = \text{mex}(1, 0, 2) = 3$
- $\text{mex}(A_2, A_3, A_4, A_5) = \text{mex}(1, 0, 2, 1) = 3$
- $\text{mex}(A_2, A_3, A_4, A_5, A_6) = \text{mex}(1, 0, 2, 1, 4) = 3$
- $\text{mex}(A_3) = \text{mex}(0) = 1$
- $\text{mex}(A_3, A_4) = \text{mex}(0, 2) = 1$
- $\text{mex}(A_3, A_4, A_5) = \text{mex}(0, 2, 1) = 3$
- $\text{mex}(A_3, A_4, A_5, A_6) = \text{mex}(0, 2, 1, 4) = 3$
- $\text{mex}(A_4) = \text{mex}(2) = 0$
- $\text{mex}(A_4, A_5) = \text{mex}(2, 1) = 0$
- $\text{mex}(A_4, A_5, A_6) = \text{mex}(2, 1, 4) = 0$
- $\text{mex}(A_5) = \text{mex}(1) = 0$
- $\text{mex}(A_5, A_6) = \text{mex}(1, 4) = 0$
- $\text{mex}(A_6) = \text{mex}(4) = 0$

Sample Input 3

25
2 2 1 2 2 0 2 2 2 1 2 2 2 0 2 2 2 2 1 2 2 2 2 0 2

Sample Output 3

624

Hints: AtCoder Beginner Contest 470

Taken from the official editorial, not written by us. Read one hint at a time.

A. Fizz *(editorial, first line)*

Use a for statement to print N lines.

B. Monocolor *(editorial, first line)*

It is optimal to paint them with the most frequent color among the current N balls.

C. Inc, Dec, Xor *(editorial, first line)*

Let $f(A)$ be the total XOR of A .

D. Inverse and Swap *(editorial, first line)*

By identifying a permutation P with a (one-to-one)

mapping $\begin{pmatrix} 1 & 2 & \dots & N \\ P_1 & P_2 & \dots & P_N \end{pmatrix}$, a query of type 2 can be seen as a replacement of ...

E. Concentration *(editorial, first line)*

Let us call an iteration (flip two cards, and gain score or lose life) a turn.

F. Googol Swaps *(editorial, first line)*

Define an undirected graph G with N vertices and M edges, whose vertices represent the positions of characters and each edge (A_i, B_i) represent the choice of i in the operation.

G. ΣΠΧ *(editorial, first line)*

Let S_k be the set of intervals where all integers between 0 and k occur at least once.

Source: <https://atcoder.jp/contests/abc470/editorial>